

1.2 Classwork: Arithmetic Sequences with Variables

1. Consider the arithmetic sequence

83, 77, 71, 65, ...

- (a) Write down the value of the common difference d .
- (b) Find an expression for u_n , the n^{th} term.
- (c) Find u_{20} .
- (d) One term of the sequence is equal to -13 . Find the value of n for that term. Show your algebraic working.

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2. The first three terms of an arithmetic sequence are

$$u_1 = k - 4, \quad u_2 = 2k, \quad u_3 = 4k - 1.$$

- Find an expression, in terms of k , for $u_2 - u_1$ and an expression for $u_3 - u_2$. Simplify your answers.
- Use the fact that the sequence is arithmetic to find the value of k .
- Write down the first three terms of the sequence.
- Find u_{10} .